The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Hessian Minimization Law
from a Centered Hidden-Response
Residual Law

Abstract

The current active-primary theorem already proves that the current canonical square-root operator core is already the canonical reduced readout of a deeper hidden-response Hessian minimization law on the fixed displacement fibers. The next honest benchmark-facing move is therefore no longer another replay of the square-root operator core, of the normalized-lift package, of the selected-lift package, of the body-level forcing law, or of the visible-response structures that follow from them. It is to derive or justify the hidden-response Hessian minimization law itself from still more primitive explicit substrate structure.

The present note takes that step in the smallest residual form currently available below the hidden-response-Hessian-minimization frontier. For each sector, it introduces a benchmark-faithful centered hidden-response residual law: the fixed proto-section benchmark base and fixed section-centered displacement body together with a finite-dimensional centered residual state space, a smooth surjective displacement readout to the actual displacement fibers, and a smooth injective centered residual operator into a Euclidean residual target. The current hidden-response Hessian minimization law is then no longer primitive. It is canonically induced by the pullback residual energy

$$A_\sigma^{\mathrm{disp,hid}}(y) = L_\sigma^{\mathrm{disp,hid,res}}(y)^* L_\sigma^{\mathrm{disp,hid,res}}(y), \qquad \mathcal{E}_\sigma^{\mathrm{hid,min}}(y,v) = \min_{\Pi_\sigma^{\mathrm{disp,hid,res}}(y)u=v} \frac{1}{2}\left\|L_\sigma^{\mathrm{disp,hid,res}}(y)u\right\|^2.$$

The benchmark boundary remains unchanged throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. Once this smaller centered residual law is fixed, the current hidden-response Hessian minimization law is forced, and therefore the same square-root operator core, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, the same current body-response core, and every currently recorded downstream benchmark-facing consequence follow unchanged. The remaining honest burden therefore moves below the separate hidden-response-Hessian package itself to a deeper derivation or justification of the centered hidden-response residual law that forces it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has already removed the larger stopping points that once sat above the current section-centered hidden-response datum.

The visible-response operator family was compressed to the body-level forcing law, the body-level forcing law was compressed to the selected-lift package, the selected-lift package was compressed to the canonical normalized selected-response lift, the normalized-lift package was compressed to the square-root operator core, and the square-root operator core was then shown to be a canonical consequence of a deeper hidden-response Hessian minimization law.

The next honest question is therefore narrower again. What still more primitive explicit substrate structure forces that current hidden-response Hessian minimization law itself? The present note answers that question in the smallest residual form now available below the current frontier. It does not reinstall the square-root operator core, the normalized-lift package, the selected-lift package, the body-law presentation, the visible-response operator family, the canonical minimal visible-response representative, the current body-response core, or any larger current-image-core or reduced-carrier or projection/body or full substrate-law presentation as independently primitive data. Instead it keeps only a centered residual state space, a displacement readout, and a residual operator into a Euclidean residual target. The current hidden-response Hessian minimization law is then the induced pullback least-squares law on that smaller residual package.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim a completed first-principles derivation of GR from final substrate microphysics. Its narrower positive role is to show that the current hidden-response Hessian minimization law already follows canonically from a smaller centered hidden-response residual law on the fixed displacement fibers.

2. Fixed Inputs from the Current Hidden-Response-Hessian Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Definition 2 (Recorded benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active route already fixes the following data and consequences.*

1. *the current proto-section benchmark base and in particular the recorded proto-section body* $\mathcal{B}^{\mathrm{psec}}_{\sigma}$ *and shell* $\mathcal{Z}^{\mathrm{psec}}_{\sigma}$*;*

2. *the fixed section-centered displacement body*

$$\mathcal{D}^{\mathrm{pppresec}}_{\sigma} \subset \mathcal{B}^{\mathrm{psec}}_{\sigma} \times V^{\mathrm{pppresec}}_{\sigma}$$

   whose fiber

$$\mathcal{D}^{\mathrm{pppresec}}_{\sigma}(y) \equiv \{v \in V^{\mathrm{pppresec}}_{\sigma} : (y, v) \in \mathcal{D}^{\mathrm{pppresec}}_{\sigma}\}$$

   is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}^{\mathrm{psec}}_{\sigma}$*;*

3. *the recorded downstream continuation*

$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,res}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec}},$$
$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,sec}},$$

   together with the current continuity-Hessian compressed core and benchmark one-metric observer package;

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Benchmark-faithful hidden-response Hessian minimization law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* benchmark-faithful hidden-response Hessian minimization law *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*

2. *A finite-dimensional Euclidean hidden-response carrier*

$$\mathcal{H}^{\mathrm{disp,hid}}_{\sigma}.$$

3. *A smooth family of surjective linear hidden-response readouts*

$$\Pi^{\mathrm{disp,hid}}_{\sigma}(y) : \mathcal{H}^{\mathrm{disp,hid}}_{\sigma} \to V^{\mathrm{pppresec}}_{\sigma}, \qquad y \in \mathcal{B}^{\mathrm{psec}}_{\sigma}.$$

4. *A smooth family of positive self-adjoint hidden Hessians*

$$A^{\mathrm{disp,hid}}_{\sigma}(y) : \mathcal{H}^{\mathrm{disp,hid}}_{\sigma} \to \mathcal{H}^{\mathrm{disp,hid}}_{\sigma}, \qquad y \in \mathcal{B}^{\mathrm{psec}}_{\sigma},$$

   each of which is positive definite on $\mathcal{H}^{\mathrm{disp,hid}}_{\sigma}$.

5. *Benchmark faithfulness in the sense of Definition 2(4).*

Proposition 1 (What the current hidden-response-Hessian frontier already proves)**.** *At current scope, the already-recorded hidden-response-Hessian frontier theorems prove all of the following.*

1. *Any benchmark-faithful hidden-response Hessian minimization law canonically determines the current square-root operator core on the fixed displacement fibers.*

2. *That square-root operator core canonically determines the current section-centered displacement-body quadratic forcing law on the fixed displacement body.*

3. *That forcing law canonically determines the visible-response operator family* $S_\sigma^{\mathrm{disp,resp,vis}}$ *and the canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$.

4. *The current body-response core and every currently recorded downstream benchmark-facing consequence then follow unchanged.*

Proof. Item (1) is exactly the content of the current theorem proving that the canonical square-root operator core is the reduced readout of a hidden-response Hessian minimization law. Items (2)–(4) are exactly the previously recorded downstream continuation from that square-root operator core. □

3. Centered Hidden-Response Residual Law

Definition 4 (Benchmark-faithful centered hidden-response residual law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* benchmark-faithful centered hidden-response residual law *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*

2. *A finite-dimensional Euclidean centered residual state space*

$$\mathcal{U}_\sigma^{\mathrm{disp,hid,res}}.$$

3. *A smooth family of surjective linear displacement readouts*

$$\Pi_\sigma^{\mathrm{disp,hid,res}}(y) : \mathcal{U}_\sigma^{\mathrm{disp,hid,res}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

4. *A finite-dimensional Euclidean residual target*

$$\mathcal{E}_\sigma^{\mathrm{disp,hid,res}}.$$

5. *A smooth family of injective linear centered residual operators*

$$L_\sigma^{\mathrm{disp,hid,res}}(y) : \mathcal{U}_\sigma^{\mathrm{disp,hid,res}} \to \mathcal{E}_\sigma^{\mathrm{disp,hid,res}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

6. *Benchmark faithfulness in the sense of Definition 2(4).*

Remark 1. *Definition 4 is intentionally smaller than the current hidden-response Hessian minimization law. It does not install a positive hidden Hessian as an independent datum. It keeps only a centered residual state space, the displacement readout, and a centered residual operator into a Euclidean residual target. The current hidden-response Hessian will be induced from that residual operator as a pullback residual norm rather than inserted separately. It also does not reinstall the square-root operator core, the normalized-lift package, the selected-lift package, the body-level forcing law, the visible-response operator family, the canonical minimal visible-response representative, the current body-response core, or any larger current-image-core or reduced-carrier or projection/body or full substrate-law presentation as primitive data.*

4. Induced Hidden-Response Hessian Minimization Law

Definition 5 (Residual-law-induced hidden-response package)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful centered hidden-response residual law. Define the induced hidden-response data by*

$$\mathcal{H}_\sigma^{\mathrm{disp,hid}} \equiv \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}, \qquad \Pi_\sigma^{\mathrm{disp,hid}}(y) \equiv \Pi_\sigma^{\mathrm{disp,hid,res}}(y), \qquad A_\sigma^{\mathrm{disp,hid}}(y) \equiv L_\sigma^{\mathrm{disp,hid,res}}(y)^* L_\sigma^{\mathrm{disp,hid,res}}(y).$$

Also define the induced least-squares residual energy on the actual displacement body by

$$\mathcal{J}_\sigma^{\mathrm{disp,hid,res}}(y,v) \equiv \min\left\{\frac{1}{2}\left\|L_\sigma^{\mathrm{disp,hid,res}}(y)u\right\|^2 : u \in \mathcal{U}_\sigma^{\mathrm{disp,hid,res}},\ \Pi_\sigma^{\mathrm{disp,hid,res}}(y)u = v\right\} \tag{2}$$

for every $(y,v) \in \mathcal{D}_\sigma^{\mathrm{ppresec}}$.

Proposition 2 (The pullback residual operator induces a positive hidden Hessian)**.** *Under Definition 5, the family* $A_\sigma^{\mathrm{disp,hid}}(y)$ *is positive self-adjoint and positive definite on* $\mathcal{H}_\sigma^{\mathrm{disp,hid}}$ *for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$.

Proof. Self-adjointness is immediate from the definition

$$A_\sigma^{\mathrm{disp,hid}}(y) = L_\sigma^{\mathrm{disp,hid,res}}(y)^* L_\sigma^{\mathrm{disp,hid,res}}(y).$$

For any nonzero $u \in \mathcal{H}_\sigma^{\mathrm{disp,hid}} = \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$,

$$\left\langle u, A_\sigma^{\mathrm{disp,hid}}(y)u\right\rangle = \left\langle L_\sigma^{\mathrm{disp,hid,res}}(y)u, L_\sigma^{\mathrm{disp,hid,res}}(y)u\right\rangle = \left\|L_\sigma^{\mathrm{disp,hid,res}}(y)u\right\|^2.$$

Because $L_\sigma^{\mathrm{disp,hid,res}}(y)$ is injective, the last quantity is strictly positive whenever $u \neq 0$. Hence $A_\sigma^{\mathrm{disp,hid}}(y)$ is positive definite. □

Proposition 3 (Residual-law-induced hidden-response Hessian minimization law)**.** *Under Definition 5, the data*

$$\left(\mathcal{H}_\sigma^{\mathrm{disp,hid}}, \Pi_\sigma^{\mathrm{disp,hid}}, A_\sigma^{\mathrm{disp,hid}}\right)$$

form a benchmark-faithful hidden-response Hessian minimization law in the sense of Definition 3.

Proof. The carrier $\mathcal{H}_\sigma^{\mathrm{disp,hid}} = \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$ is finite-dimensional and Euclidean. The readout family $\Pi_\sigma^{\mathrm{disp,hid}}(y) = \Pi_\sigma^{\mathrm{disp,hid,res}}(y)$ is smooth and surjective by Definition 4(3). Proposition 2 shows that the hidden Hessian family is smooth, self-adjoint, and positive definite. Benchmark faithfulness is inherited directly from Definition 4(6). Therefore all items of Definition 3 hold. □

Proposition 4 (The current hidden-response minimization law is exact least-squares residual energy)**.** *Under Definition 5, for every* $(y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}$,

$$\mathcal{J}_\sigma^{\mathrm{disp,hid,res}}(y, v) = \min \left\{ \frac{1}{2} \left\langle u, A_\sigma^{\mathrm{disp,hid}}(y) u \right\rangle : u \in \mathcal{H}_\sigma^{\mathrm{disp,hid}},\ \Pi_\sigma^{\mathrm{disp,hid}}(y) u = v \right\}. \tag{3}$$

Proof. For every admissible u,

$$\frac{1}{2} \left\langle u, A_\sigma^{\mathrm{disp,hid}}(y) u \right\rangle = \frac{1}{2} \left\langle u, L_\sigma^{\mathrm{disp,hid,res}}(y)^* L_\sigma^{\mathrm{disp,hid,res}}(y) u \right\rangle = \frac{1}{2} \left\| L_\sigma^{\mathrm{disp,hid,res}}(y) u \right\|^2.$$

Because the readout families agree by Definition 5, the admissible sets in (2) and (3) are the same. Therefore the two minimization problems coincide exactly. □

5. Deriving the Current Hidden-Response-Hessian Frontier

Theorem 1 (Current hidden-response Hessian minimization law from a centered hidden-response residual law)**.** *Fix the primitive continuation data of Definition 1, the recorded benchmark base and downstream chain of Definition 2, and for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *a benchmark-faithful centered hidden-response residual law in the sense of Definition 4. Then the current hidden-response Hessian minimization law is canonically forced by that smaller explicit substrate package. More precisely:*

1. *Proposition 3 canonically associates to the centered residual law a benchmark-faithful hidden-response Hessian minimization law* $(\mathcal{H}_\sigma^{\mathrm{disp,hid}}, \Pi_\sigma^{\mathrm{disp,hid}}, A_\sigma^{\mathrm{disp,hid}})$.

2. *Proposition 4 identifies the current hidden-response minimization law exactly with the least-squares residual law*

$$\mathcal{E}_\sigma^{\mathrm{hid,min}}(y, v) = \mathcal{J}_\sigma^{\mathrm{disp,hid,res}}(y, v) = \min_{\Pi_\sigma^{\mathrm{disp,hid,res}}(y) u = v} \frac{1}{2} \left\| L_\sigma^{\mathrm{disp,hid,res}}(y) u \right\|^2.$$

3. *By Proposition 1, the current canonical square-root operator core, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family* $S_\sigma^{\mathrm{disp,resp,vis}}$, *the same canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$, *the same current body-response core, and every currently recorded downstream benchmark-facing consequence then follow unchanged from that same centered residual law through the induced hidden-response Hessian minimization law.*

4. *The benchmark one-metric package* (1) *is preserved, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

5. *Consequently the remaining honest burden is no longer to restate the current hidden-response Hessian minimization law itself, but to derive or justify the centered hidden-response residual law that forces it from still more primitive explicit substrate structure.*

Proof. Item (1) is Proposition 3. Item (2) is Proposition 4. Item (3) is the already-recorded downstream continuation of the current hidden-response-Hessian frontier applied to the law furnished by Items (1)–(2). Item (4) is the benchmark-faithfulness condition built into Definitions 2 and 4. Item (5) is the routing consequence of Items (1)–(4): once the current hidden-response Hessian

minimization law is shown to be a canonical pullback least-squares law of the smaller centered residual package, another manuscript that merely restates that same minimization law or merely replays the same square-root-core or normalized-lift or selected-lift or body-law or visible-response consequences does not remove any remaining benchmark-facing freedom. □

Corollary 1 (The hidden-response Hessian minimization law is no longer primitive)**.** *Under the hypotheses of Theorem 1, the current hidden-response Hessian minimization law is no longer a primitive stopping point on the recorded section-centered hidden-response route. Its benchmark-relevant content is already forced by the smaller centered hidden-response residual law* $\mathfrak{N}^{\mathrm{disp,hid,res}}$*.*

Proof. Theorem 1(1)–(3) identifies the current hidden-response Hessian minimization law and its same square-root-core and downstream benchmark-facing consequences as outputs of the smaller centered hidden-response residual law. Therefore the current hidden-response-Hessian package is no longer primitive at benchmark scope once that residual law is fixed. □

6. Conclusion

The positive result of this note is a genuine derivational gain below the current hidden-response-Hessian frontier. The current hidden-response Hessian minimization law no longer needs to be treated as primitive. It is already the canonical pullback least-squares law of a smaller centered hidden-response residual law on the fixed displacement fibers.

The scope remains narrow and honest. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and nothing here upgrades adoption to a completed first-principles derivation of GR from final substrate microphysics.

The next open burden is therefore sharper again. The next benchmark-facing manuscript should derive or justify the centered hidden-response residual law itself from still more primitive explicit substrate structure. Another manuscript that merely restates the same centered residual law, the same hidden-response Hessian minimization law, the same square-root operator core, the same carrier-bearing normalized selected-response lift, the same operator-bearing selected-lift package, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, the same current body-response core, or a larger same-output current-image-core or reduced-carrier or projection/body or full substrate-law presentation is no longer the clean next target at current scope.